

Banking Analytics Dashboard

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Data Analytics · Power BI · 2026

Introduction & Objectives

What is this project?

A complete Banking Analytics Dashboard built on real banking data covering 5,000 customers from 2021 to 2024.

*"This bank is silently losing both its customers and its revenue at the same time.
Let me show you exactly what the data is telling us."*

Four Business Questions Answered:

-  Sales & Revenue → Which segment & branch drives most revenue? What is ARPC?
-  Churn & Retention → What % of customers churned? What is causing churn?
-  Operations & Risk → Which branches show high NPA? What is EMI stress level?
-  Risk & Fraud → Where are fraud losses highest? What is recovery rate?

Dataset Overview

Table	Records	Fields
Customers	5,000	20
Accounts	3,550	19
Transactions	3,230	18
Loans	2,743	20
Fraud Alerts	3,543	17
TOTAL	20,066	94

Period → 2021 to 2024 Domain → Banking & Finance

Relationship → Customers → Accounts → Transactions | Customers → Loans | Transactions → Fraud Alerts

Data Cleaning Process

(Power Query — Step by Step)

What Was Wrong With the Data:

Issue 1 → Handled duplicate records across multiple tables.

Issue 2 → Missing values — nulls in 80% of columns

Issue 3 → Detected spelling inconsistencies, mixed text formatting, duplicate records, and inconsistent date structures across multiple tables.

Issue 4 → 3+ date formats mixed in every single table

Issue 5 → Mixed case text — "verified" / "Verified" / "VERIFIED" — all same thing

How I Fixed It:

Step 1 → Removed 535 duplicate rows across all 5 tables

Step 2 → Standardized text — Trim + Capitalize Each Word + Replace Values

Step 3 → Fixed all date formats — 3+ formats unified to single Date type

Step 4 → Filled nulls with business logic (close_date null = Still Open | is_flagged null = 0)

Data Model & Relationships — Star Schema

[CUSTOMERS]

Center Table · All connected through customer_id

Accounts

Transactions

Loans

Fraud Alerts

4 Relationships → All Many-to-One → Tables were connected using customer_id and transaction-level relationships for integrated analysis.

Why Star Schema?

- Customer-level relationships enabled integrated cross-functional banking analysis.
- One slicer (Year / Segment / City) updates all charts on all pages instantly
- Clean Many-to-One relationships make DAX measure calculation reliable
- Optimized structure for fast cross-table analysis in Power BI

Dashboard Pages Overview

Page 1 → Sales & Revenue

4 KPI Cards + 4 Charts → 4 business questions answered

Page 2 → Customer Churn & Retention

5 KPI Cards + 4 Charts → 4 business questions answered

Page 3 → Operations & Risk Analysis

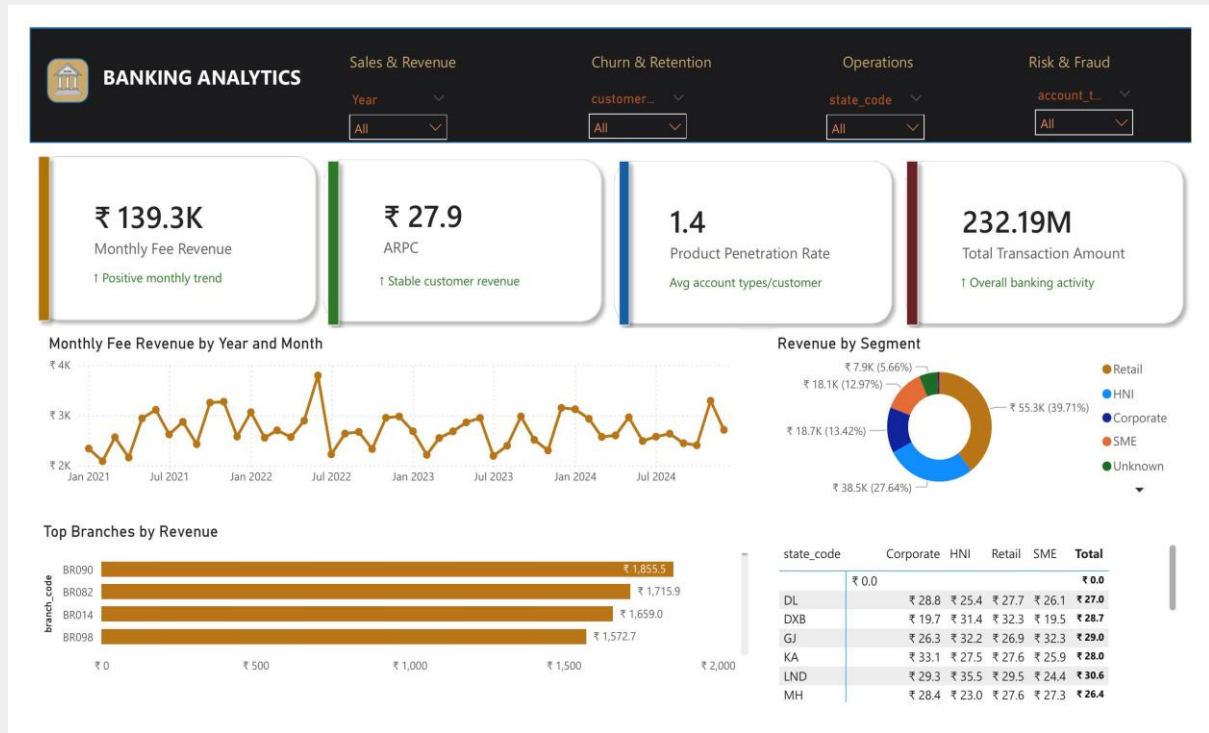
5 KPI Cards + 4 Charts → NPA, EMI stress, overdue loans , emi stress distribution

Page 4 → Risk & Fraud Detection

5 KPI Cards + 4 Charts → Fraud loss, recovery, hotspots, Fraud severity impact analysis

Tools Used → Power BI Desktop · Power Query (cleaning) · DAX (25 measures) · Star Schema (data model)

Page 1 — Sales & Revenue Dashboard



The line chart shows monthly revenue trend from 2021 to 2024. The donut chart shows Retail segment leading at 39.71%. The bar chart shows BR090 as the top branch with ₹1,855. And the table on the right shows LND region has the highest ARPC at ₹30.6.”

What Revenue Dashboard Revealed

Finding 1 → Revenue Decline

Total fee revenue dropped 35% — from ₹1.4K (2021) to ₹0.9K (2024)

Finding 2 → Segment Paradox

Retail and HNI segments contributed significantly to overall revenue generation.

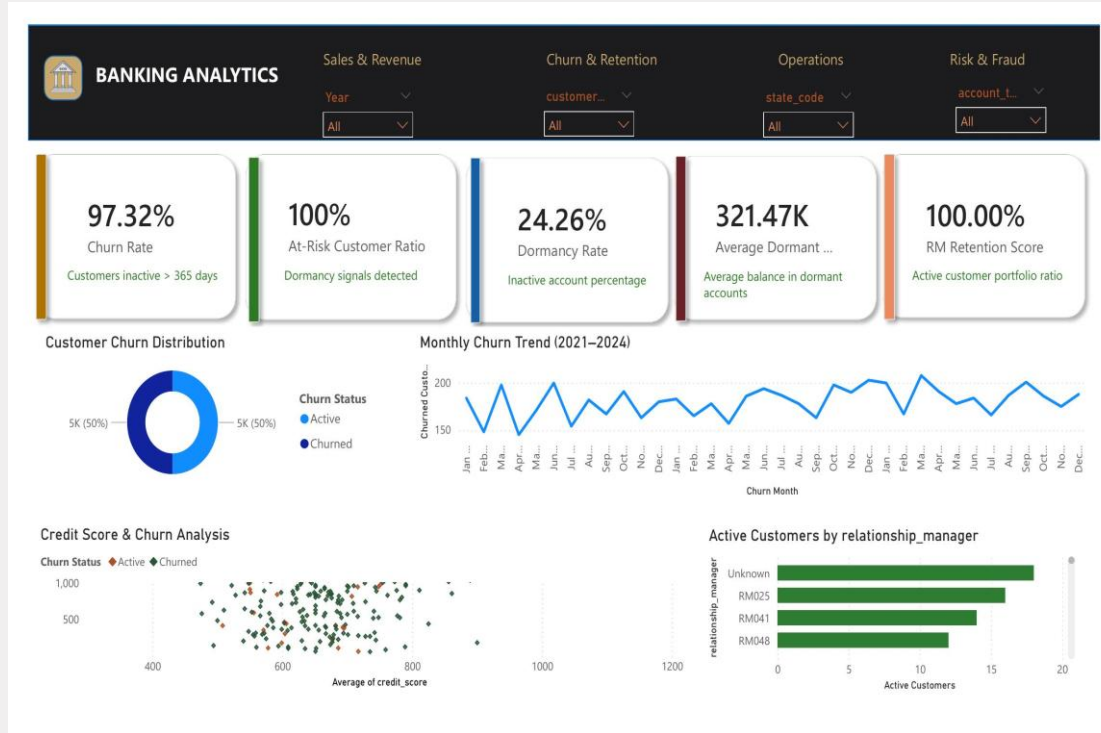
Finding 3 → Branch Gap

Revenue contribution varied significantly across branches, with BR090 leading overall performance.

Finding 4 → City ARPC Inequality

LND region recorded the highest ARPC among tracked regions.

Page 2 — Customer Churn & Retention Dashboard



The donut chart shows 50% active and 50% churned. The monthly trend shows churn rising between 150 and 200 per month. The scatter plot shows credit score versus churn. And RM025 and RM041 are the top relationship managers.

What Churn Dashboard Revealed

Finding 1 → Retention Crisis

A significant portion of customers were classified as churned based on inactivity rules.

Finding 2 → At-Risk Warning

A noticeable number of customers fall into the at-risk inactivity category.

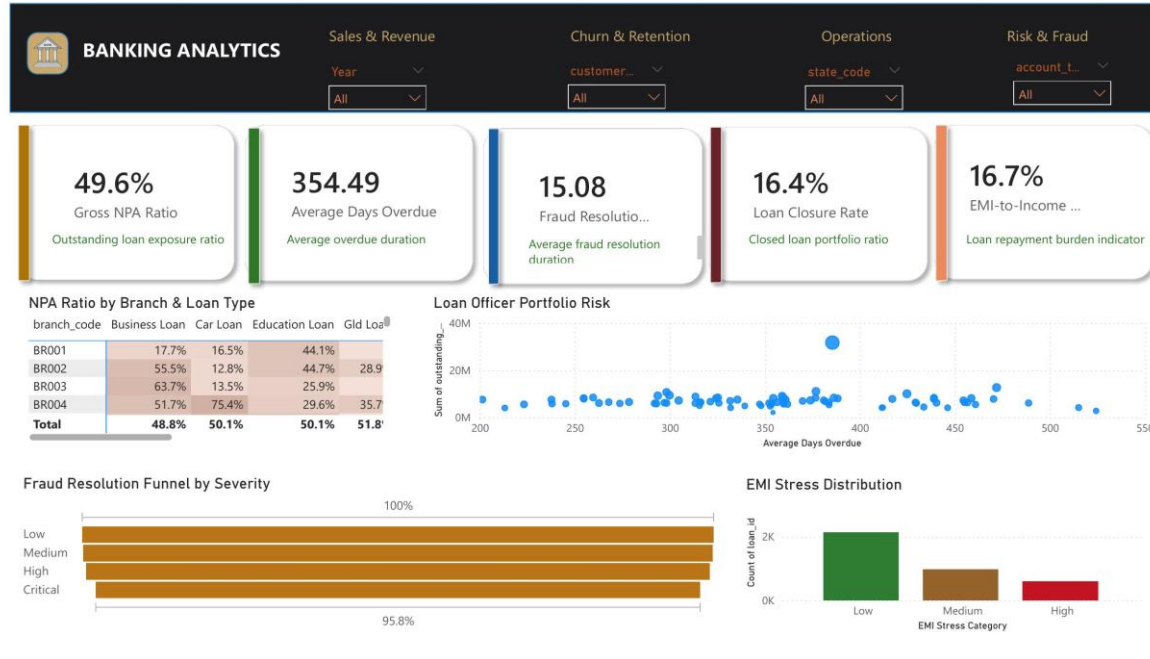
Finding 3 → Scatter Plot Result

WEAK correlation between credit score and churn — Customer engagement and inactivity patterns appeared to influence churn more strongly than credit score alone.

Finding 4 → Dormancy Pattern

Recurring Deposit has the highest dormancy rate — ₹298,970 average balance sitting idle

Page 3 — Operations & Risk Analysis Dashboard



The table on the left shows NPA by branch and loan type — BR003 has 63.7% Business Loan NPA. The scatter plot shows loan officer portfolio risk. And the EMI stress chart shows the distribution of stress categories.

What Operations Dashboard Revealed

Finding 1 → High NPA Exposure

Gross NPA ratio is 49.6% — nearly half of all outstanding loans are non-performing

Finding 2 → Worst Branch

BR003 recorded the highest Business Loan NPA exposure among observed branches.

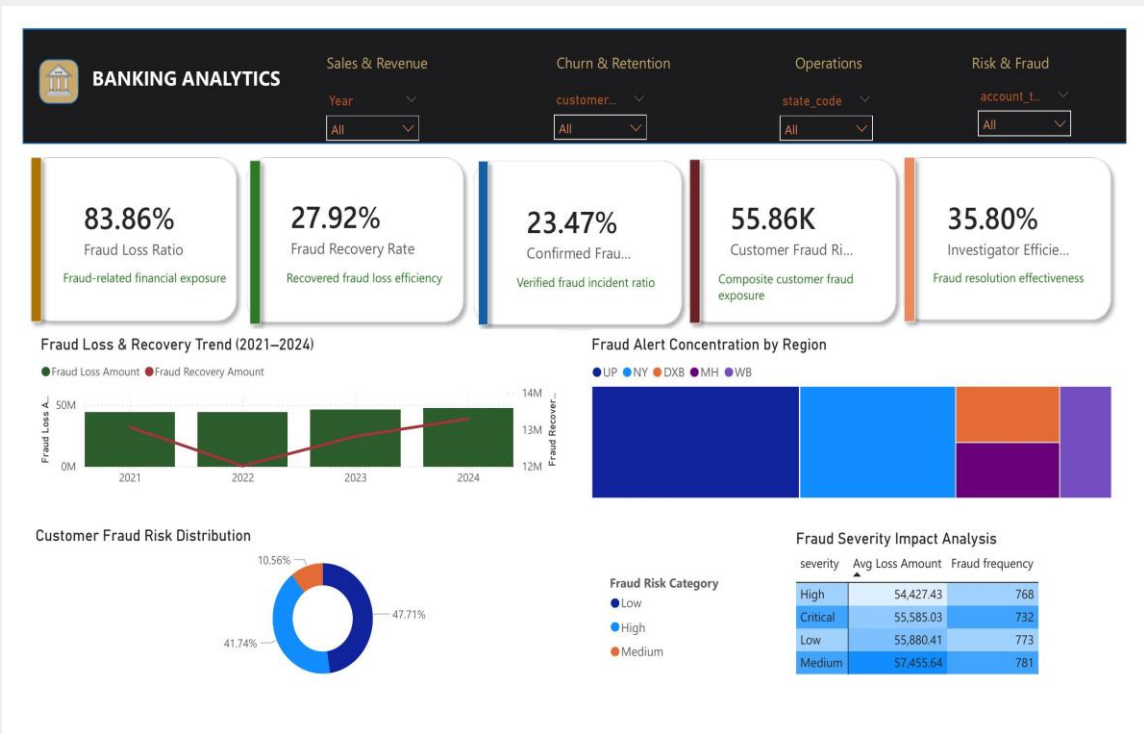
Finding 3 → Overdue Duration

Average days overdue = 354 days — loans are unpaid for almost a full year on average

Finding 4 → EMI Stress Risk

980 accounts are in High EMI-stress category — most likely to default if no intervention

Page 4 — Risk & Fraud Detection Dashboard



The trend chart shows fraud losses of around 50 million per year versus recovery of 12 to 14 million. The region chart shows UP has the highest fraud concentration. And the severity table shows Medium fraud has the highest average loss at ₹57,455.

What Fraud Dashboard Revealed

Finding 1 → Fraud Loss is Huge

Fraud recovery performance remained significantly lower than estimated fraud losses.

Finding 2 → Recovery Rate is Low

Only 27.92% of fraud losses are recovered — losses remain ~₹50M per year

Finding 3 → Geographic Hotspots

UP and Maharashtra have the highest fraud alert concentration in the country

Finding 4 → Medium Severity Worst

Medium severity frauds carry the highest avg loss — ₹57,455 per case (more than Critical)

Key Findings — Summary



Revenue

35% fee revenue decline · 60× ARPC gap · 2.3× branch performance gap



Churn

Significant ARPC variation across regions · Churn = SERVICE problem not financial



Operations

49.6% Gross NPA · 354 avg overdue days · BR003 = 63.7% NPA worst branch



Fraud

83.86% fraud loss ratio · UP & MH hotspots · Medium severity highest loss

"The opportunity for recovery is ₹213M+ sitting right inside our existing customer base."

Business Recommendations

1. Focus on HNI & Corporate segments in high-ARPC cities like Delhi to maximize revenue
2. Study and replicate top-performing branch strategies across lower-performing branches.
3. Launch early churn detection at 180-day inactivity threshold — before the 1-year mark
4. Implement segment-specific re-engagement campaigns — churn is a service problem
5. Urgently review BR003's NPA portfolio — 63.7% Business Loan NPA is unsustainable
6. Offer EMI restructuring to 980 high-stress accounts before they default
7. Deploy fraud investigation teams specifically in UP and Maharashtra regions
8. Prioritize Medium-severity fraud cases — they carry the highest financial loss per case
9. Activate dormant Recurring Deposit accounts — ₹298K average idle balance is recoverable

Conclusion

Our dashboard has told us a complete story.

A significant number of customers showed prolonged inactivity patterns..

A bank processing ₹232.19M in transactions is watching fee income trends showed periods of decline during the analysis period on year.

significant revenue variation across regions and customer segments.

The analysis highlights clear opportunities for improving customer retention, operational efficiency, and fraud recovery performance within the existing customer base. The greatest growth opportunity lies not in acquiring more customers, but in strengthening the relationships we already have.

Developed using Power BI · DAX · Power Query · Star Schema Data Modeling · 20,066 Records · 25 DAX Measures

Thank You

"Behind every data point lies a customer journey, a business decision, and an insight powerful enough to drive change."

Tool Used → Microsoft Power BI & Excel

Dataset → Banking & Finance

Period → 2021 – 2024

Records → 20,066 rows analysed

Measures → 25 DAX Measures